



Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Mathematics

Graphical Analysis & Differential Calculus

Part A

All questions are valued at a total of two marks, unless otherwise specified.

	<i>Marks</i>
1. Find the radius of the curvature at the point $(2, 8)$ on the curve $y = 2x^2$.	(2)

	<i>Marks</i>
2. Find the radius of the curvature, ρ , for $y = \frac{1}{6}x^3$ at the point $(2, \frac{4}{3})$.	(2)

	<i>Marks</i>
3. Find the radius of the curvature for the graph of f at the point $(1, 3)$, where $f(x) = (x - 1)^2 + 3$.	(2)

	<i>Marks</i>
4. If $y = x + 3x^2 - x^3$, find the radius of the curvature at origin.	(2)

	<i>Marks</i>
5. For the curve defined by $y = \frac{x(1-x+x^2)}{2}$, find the radius of the curvature at the point $(1, \frac{1}{2})$.	(2)

		<i>Marks</i>
6.	Given that a is an element of real numbers, find the radius of the curvature, ρ , at the point $(a, 0)$ on the curve $y = x^3(x - a)$.	(2)

Part B

All questions are valued at a total of six marks, unless otherwise specified.

		Marks
1.	(i) Define the term “radius of curvature.”	(2)
	(ii) Find the radius of the curvature for $y = \frac{1}{x^2+1}$ at the points given at $x = 0$ and $x = 2$ and hence determine which of these points has a greater curvature.	(4)

		Marks
2.	(i) Explain the relationship between curvature and radius of curvature, and hence write down the Cartesian formula for curvature, κ , at a point on a curve.	(2)
	(ii) Find the radius of the curvature at the points $P(0, \sqrt{1})$ and $Q(1, 0)$ on the curve $y = \sqrt{1 - x^2}$ and hence determine which of these points has a greater curvature.	(4)

Prepared By

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Prepared Independently Dr. R. Williams to meet the requirements of Algebra and Calculus course at SSN College of Engineering. This text written by him is not endorsed by SSN College of Engineering, Af. Anna University.